

Muhammad Hasnain

Full Stack Software Engineer | AI Agent Developer | Prompt Engineer

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PROFESSIONAL SUMMARY

Full Stack Software Engineer with 4+ years of experience building scalable web applications using .NET Core, Angular, and SQL Server. Specializes in RESTful API development, real-time systems with SignalR, and containerized deployments with Docker. Recently developed a production AI Agent using the OpenAI API for automated bug detection and resolution. Combines strong backend and frontend engineering with prompt engineering and LLM integration skills to deliver intelligent, modern software solutions. Experienced with Azure DevOps CI/CD pipelines and Agile/Scrum methodologies.

TECHNICAL SKILLS

AI & LLM: OpenAI API, Prompt Engineering, AI Agent Development, LLM Integration

Languages: C#, TypeScript, JavaScript, SQL

Frontend: Angular, React, Tailwind CSS, Material UI, HTML5, CSS3

Backend: .NET Core, Node.js, Express.js, REST API, RESTful APIs

Databases: SQL Server, MySQL, MongoDB, Redis

Real-Time & Messaging: SignalR, Business WhatsApp API

DevOps & Cloud: Docker, Azure DevOps, CI/CD Pipelines, AWS, Heroku, Netlify

Security & Auth: JWT Authentication, Role-Based Access Control (RBAC), ORM

Methodologies: Agile, Domain-Driven Design (DDD), SOLID Principles, OOP

Tools: Git, Postman, Visual Studio, VS Code, SyncFusion, DevExtreme, Trello, Asana

PROFESSIONAL EXPERIENCE

Full Stack Software Engineer

04/2025 - Present

DMCR (DM Clinical Research) | Remote - Houston, Texas

- Engineered an AI Agent using the OpenAI API that autonomously detects production bugs, analyzes root causes, and generates ready-to-apply code solutions - significantly reducing mean time to resolution (MTTR).
- Applied advanced prompt engineering techniques to optimize LLM outputs for code analysis, improving accuracy and relevance of AI-generated bug fixes.
- Working on ATOM, a centralized clinical trial management platform built with .NET Core, Angular, and SQL Server.
- Revamped platform UI/UX and restructured core architecture to improve scalability, maintainability, and performance.
- Identified and resolved critical production bugs, significantly enhancing system stability and reliability.
- Implemented and managed Azure DevOps CI/CD pipelines for automated builds, testing, and deployments across multiple environments.
- Actively worked with Azure Sprints — managing sprint boards, tracking user stories, updating task progress, and contributing to end-to-end Agile delivery cycles.

Software Engineer

05/2023 - 04/2025

Punjab Healthcare Commission | Lahore, Pakistan

- Developed full-stack applications using Angular, .NET Core, and SQL Server to support healthcare inspections and complaint management workflows.
- Integrated Business WhatsApp API for automated complaint handling, improving response time and user accessibility.
- Leveraged a self-built AI Agent (OpenAI API) to analyze and optimize SQL queries and stored procedures, reducing data retrieval time from 1.4 minutes to 4.5 seconds - a 96% performance improvement across high-traffic modules.
- Enhanced PHC Online Application with new validation logic, security features, and performance improvements.

- Built admin dashboards with interactive charts and graphs using SyncFusion, enabling data-driven decision-making.

Software Engineer

08/2022 - 04/2023

Lab Code | Bahawalpur, Pakistan

- Built Ed-Tech and FinTech solutions using the MERN stack (MongoDB, Express.js, React, Node.js).
- Designed and developed RESTful APIs with JWT authentication and authorization middleware for secure access control.
- Optimized MongoDB queries, improving system efficiency and reducing response times.

Junior Web Developer

01/2021 - 07/2022

Techticks Solutions | Bahawalpur, Pakistan

- Built and maintained web applications using JavaScript, React, and Node.js.
- Debugged and resolved performance bottlenecks, improving application responsiveness.

KEY PROJECTS

AI Bug Detection Agent | *OpenAI API, .NET Core, Prompt Engineering, Azure DevOps*

- Designed and deployed an AI Agent at DMCR that monitors production systems, autonomously identifies bugs, and generates detailed solutions with code suggestions.
- Engineered prompt pipelines to feed error logs and codebase context to the LLM, producing context-aware, actionable resolutions.
- Reduced developer debugging time by automating root-cause analysis and solution drafting — directly contributed to a 96% SQL query performance improvement (1.4 min to 4.5 sec) at DMCR.

ATOM - Clinical Trial Inspection Platform | *Angular, .NET Core, SQL Server, Azure DevOps*

- Led Angular frontend setup, integrated .NET Core REST APIs, and engineered SQL stored procedures, views, and triggers for reliable data flow.
- Streamlined inspection tracking, reporting, and document handling, improving operational efficiency in clinical trial oversight.

SixaTaxis Web Application | *.NET Core, SignalR, Redis, Docker, SQL Server*

- Implemented real-time ride status communication using SignalR integrated with Redis for high-performance, scalable data handling.
- Dockerized backend services for consistent deployments; conducted thorough REST API testing with Postman.

E-Diary Document Management System | *.NET Core, Angular, SQL Server*

- Developed a document management system with Role-Based Access Control (RBAC) and automated workflow routing across organizational hierarchy.

E-Service Complaint & Inspection Management | *.NET Core, Angular, SQL Server, WhatsApp API*

- Built complaint management module with WhatsApp integration and real-time admin dashboard using SyncFusion charts.

Restaurant POS System | *MERN Stack, JWT, Node.js, MongoDB*

- Built a Point-of-Sale system with JWT authentication, custom authorization middleware, and RESTful API integration.

CERTIFICATIONS

- SQL (Basic) - HackerRank
- JavaScript (Basic) - HackerRank

EDUCATION

Bachelor of Computer Systems Engineering

Islamia University of Bahawalpur, Pakistan

LANGUAGES

- English - Professional Working Proficiency
- Urdu - Native